



1 Decay Scheme

⁶⁷Cu disintegrates by β^- emissions to several levels of ⁶⁷Zn.

Le ⁶⁷Cu se désintègre par émission β^- vers plusieurs niveaux du ⁶⁷Zn.

2 Nuclear Data

$T_{1/2}(\text{}^{67}\text{Cu})$: 61,82 (5) h
 $Q^-(\text{}^{67}\text{Cu})$: 560,8 (8) keV

2.1 β^- Transitions

	Energy (keV)	Probability (%)	Nature	log <i>ft</i>
$\beta_{0,3}^-$	167,3 (8)	1,039 (14)	Allowed	5,771
$\beta_{0,2}^-$	376,2 (8)	51,7 (6)	Allowed	5,224
$\beta_{0,1}^-$	467,5 (8)	19,73 (47)	Allowed	5,965
$\beta_{0,0}^-$	560,8 (8)	27,5 (8)	Allowed	6,098

2.2 Gamma Transitions and Internal Conversion Coefficients

	Energy (keV)	P _{$\gamma+ce$} (%)	Multipolarity	α_K	α_L (10 ⁻³)	α_M (10 ⁻³)	α_T
$\gamma_{2,1}(\text{Zn})$	91,27 (1)	6,90 (11)	M1+1.5(6)%E2	0,081 (5)	8,7 (7)	1,25 (9)	0,091 (6)
$\gamma_{1,0}(\text{Zn})$	93,310 (9)	27,36 (46)	E2	0,748 (11)	92,1 (13)	13,00 (19)	0,854 (12)
$\gamma_{2,0}(\text{Zn})$	184,579 (10)	44,9 (6)	M1+8.8(36)%E2	0,0151 (19)	1,58 (20)	0,23 (3)	0,0169 (21)
$\gamma_{3,2}(\text{Zn})$	208,943 (10)	0,1077 (24)	M1+0.18(14)%E2	0,00806 (13)	0,827 (13)	0,1186 (19)	0,00901 (14)
$\gamma_{3,1}(\text{Zn})$	300,213 (10)	0,732 (13)	M1+3.07(33)%E2	0,00348 (6)	0,354 (6)	0,0508 (8)	0,00388 (6)
$\gamma_{3,0}(\text{Zn})$	393,522 (10)	0,1988 (48)	M1+0.26(16)%E2	0,001728 (25)	0,1748 (25)	0,0251 (4)	0,00193 (3)

3 Atomic Data

3.1 Zn

ω_K	:	0,486	(4)
$\bar{\omega}_L$	:	0,0108	(4)
n_{KL}	:	1,326	(4)

3.1.1 X Radiations

	Energy (keV)		Relative probability	
X _K				
	Kα ₂	8,61587	51,42	
	Kα ₁	8,63896	100	
	Kβ ₁	9,5721	}	21,46
	Kβ ₅ ''	9,6499		
X _L				
	Lℓ	0,8836		
	Lα	1,0119 - 1,0122		
	Lη	0,9065		
	Lβ	1,02044 - 1,1861		
	Lγ	1,04333 - 1,04333		

3.1.2 Auger Electrons

	Energy (keV)	Relative probability
Auger K		
KLL	7,209 - 7,554	100
KLX	8,307 - 8,631	28,3
KXY	9,387 - 9,642	2,01
Auger L	0,7479 - 1,1846	

4 Electron Emissions

		Energy (keV)		Electrons (per 100 disint.)
e _{AL}	(Zn)	0,7479 - 1,1846		17,51 (15)
e _{AK}	(Zn)			
	KLL	7,209 - 7,554	}	6,28 (14)
	KLX	8,307 - 8,631		
	KXY	9,387 - 9,642		
ec _{2,1} T	(Zn)	81,61 - 91,27		0,575 (39)
ec _{2,1} K	(Zn)	81,61 (1)		0,512 (33)
ec _{1,0} T	(Zn)	83,65 - 93,31		12,61 (26)
ec _{1,0} K	(Zn)	83,651 (9)		11,04 (24)
ec _{1,0} L	(Zn)	92,12 - 92,29		1,359 (29)
ec _{1,0} M	(Zn)	93,17 - 93,30		0,1919 (41)
ec _{2,0} T	(Zn)	174,92 - 184,578		0,75 (9)
ec _{2,0} K	(Zn)	174,92 (1)		0,67 (8)
ec _{3,2} T	(Zn)	199,284 - 208,942		0,000962 (26)
ec _{3,1} T	(Zn)	290,554 - 300,212		0,00283 (7)
ec _{3,0} T	(Zn)	383,863 - 393,521		0,000383 (11)
$\beta_{0,3}^-$	max:	167,3 (8)	}	1,039 (14)
	avg:	45,66 (24)		
$\beta_{0,2}^-$	max:	376,2 (8)	}	51,7 (6)
	avg:	114,55 (28)		
$\beta_{0,1}^-$	max:	467,5 (8)	}	19,73 (47)
	avg:	147,50 (29)		
$\beta_{0,0}^-$	max:	560,8 (8)	}	27,5 (8)
	avg:	182,54 (31)		

5 Photon Emissions

5.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)		
XL	(Zn)	0,8836 - 1,1861	0,183 (4)		
XK α_2	(Zn)	8,61587	1,77 (4)	}	K α
XK α_1	(Zn)	8,63896	3,44 (8)		
XK β_1	(Zn)	9,5721	0,737 (18)	}	K' β_1
XK β_5''	(Zn)	9,6499			

5.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma_{2,1}(\text{Zn})$	91,27 (1)	6,32 (10)
$\gamma_{1,0}(\text{Zn})$	93,310 (9)	14,76 (23)
$\gamma_{2,0}(\text{Zn})$	184,579 (10)	44,2 (6)
$\gamma_{3,2}(\text{Zn})$	208,943 (10)	0,1068 (24)
$\gamma_{3,1}(\text{Zn})$	300,212 (10)	0,729 (13)
$\gamma_{3,0}(\text{Zn})$	393,521 (10)	0,1985 (48)

6 Main Production Modes

⁶⁵Cu(2n)⁶⁷Cu

⁶⁴Ni(α ,p)⁶⁷Cu

⁶⁷Zn(n,p)⁶⁷Cu

⁶⁷Zn(d,2p)⁶⁷Cu

⁶⁸Zn(p,2p)⁶⁷Cu

⁶⁸Zn(γ ,p)⁶⁷Cu

⁶⁸Zn(α , α p)⁶⁷Cu

⁷⁰Zn(p, α)⁶⁷Cu

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